OutlineFlow

From a bare outline → a full apartment, as vector polygons

Mirror Mirror on the Wall — DAVIS × Paris 2026

Conditional layout generation on Modified Swiss Dwellings

The task

- Input: one apartment / floor OUTLINE (the only condition)
- Output: a set of TYPED ROOM POLYGONS (vector, never pixels)
- Model: diffusion or flow matching, from scratch, seed 42
- Scored on: FID (realism ↓) + Density & Coverage (diversity ↑)

Data: Modified Swiss Dwellings — full set: 5,372 plans / 203k rooms

Our model — pure flow matching

- From-scratch rectified-flow set-Transformer v(x_t, t, outline)
- Rooms are a SET → no positional encoding; full self-attention
  - learns a joint, non-overlapping, outline-filling layout
- Outline: permutation-invariant PointNet; time+outline via DiT AdaLN-Zero
- Decode clips + resolves overlaps + gap-fills → union(rooms) == outline

1.33M params · trained on all 5,372 plans

First results — and a wall

	FID ↓	Density ↑	Coverage ↑
Baseline	167	0.097	0.067
ideal	0	1.0	1.0

Geometry clean, room counts right … but Density & Coverage ≈ 0.07–0.09.

The model is NOT diverse enough — it keeps drawing the same 'average' apartment.

Diagnosis — it's the objective, not the model

L2 training regresses to the CONDITIONAL MEAN of all valid layouts.

- Many valid plans per outline → their mean is one blurred plan → variance collapse
- Generated room-count std ±6 vs real ±24 (can't do tiny or huge plans)
- Room-count ↔ area correlates at $r = 0.88$, but the encoder normalizes size away

→ This predicts what will and won't work.

Three complementary levers

Lever	Targets	Idea
1 · Grid-align (post-proc)	FID	snap rooms to building axes → clean tiling
2 · SDE / churn sampling	Coverage	inject noise → escape the mean trajectory
3 · Adversarial / EBM	Density	critic pulls samples to the real manifold

Each attacks a DIFFERENT metric. Lever 2 (churn) directly undoes the

variance collapse — no retraining needed.

Results

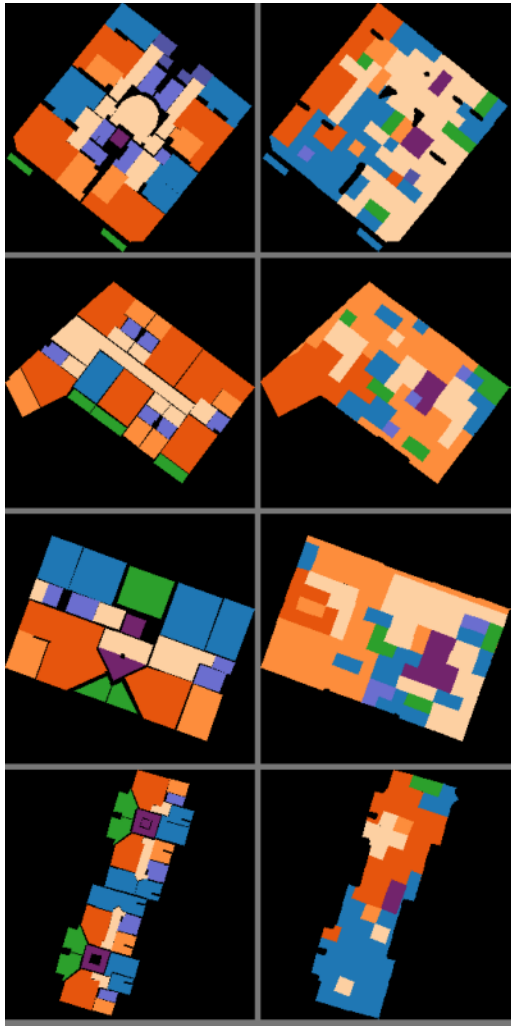
Config (3-seed mean)	FID ↓	Density ↑	Coverage ↑
Baseline	167	0.097	0.067
+ grid-align	131.5	0.091	0.098
+ churn (final)	135.3	0.088	0.111

FID −19%, Coverage +66% over the raw baseline.

Shipped as generate(outline) → vector polygons, deterministic at seed 42.

Qualitative samples

left: real plan   right: generate(outline)



We tested all three paradigms — honestly

Paradigm (align g16)	FID	Density	Coverage
Flow matching + churn	135	0.088	0.111
DDPM (pure diffusion)	137	0.068	0.090
Flow + EBM guidance	136	0.091	0.103

- Diffusion hits the SAME L2 ceiling — no better.
- EBM gains wash out under multi-seed (a churn run of 0.122 → 0.097 averaged!).

Key takeaways

- 1. The bottleneck is the LOSS, not the architecture
  - capacity, depth, paradigm swaps all fail
- 2. Sampling-time stochasticity is the real diversity lever
  - simple, robust, free
- 3. Multi-seed or it didn't happen
  - stochastic pipelines make single runs lie

Final: flow matching + grid-align + churn → FID 135 / Den 0.088 / Cov 0.111